Improving CPU Efficiency for Data-Intensive Applications through Cache-Aware Spill Optimization

Register Spill Behaviour and Cache Hit/Miss Rates on RISC-V SPEC CPU2017

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gem5 TimingSimpleCPU · workload profile `ref` · L1D 4 KiB, L1I 32 KiB, L2 32 KiB, L1D 2-way, L2 8-way · gem5 `e0a12aa4e3+dirty`

SimPoint estimates. Ref runs are too long to simulate in full: every value is estimated from up to 30 simulation points of 100 M instructions per run, each after a 500 M-instruction warmup, weighted by SimPoint (`cache_scripts/simpoint.sh`; points and weights in each run directory's `simpoint.txt`). `total_insts` is the whole program's count from qemu.

Definitions (metric schema 9). Every stack store issued inside the region of interest, and every later load that reads it back (the load's address lies in the bytes the store wrote; widths may differ), belongs to exactly one class. `spill`: register-allocator spill code, identified by GCC itself (`-dP` RTL dumps of the same build, memory operands in the LRA spill area). `csave`: callee-saved register save/restore, return address included, identified from the binary's CFI (`.eh_frame`). `other_stack`: every remaining SP/FP-relative stack store and reload (stack-passed arguments, memory-resident locals). `stack_*` is the sum of the three. For each class an *event* is one (store, first reload) pair, so `event` $\leq$ `store` and `load` $\geq$ `event`. All counters cover only the region delimited by `m5_work_begin`/`m5_work_end`: the gem5 statistics are dumped and reset at those boundaries, so the cache columns span the same window, and the cache tags carry the class, so per-class hit/miss rates come straight from gem5. *Spill %* and *Csave %* are the class's store+reload instructions over `ROI_insts`; *spill/stack %* is the share of stack events that are real spills.

Benchmark	ROI_insts	stack_event	**spill_event**	csave_event	other_event	spill/stack %	Spill %	Csave %	L1D m %	L2 m %	sp_load m %	sp_store m %	cs_load m %	cs_store m %
cactuBSSN	3 831 491 985 262	309 177 145 504	308 680 668 558	495 633 510	843 436	99.8	20.65	0.03	15.638	64.77	4.393	3.597	0.099	0.058
parest	4 088 545 717 646	14 998 028 893	811 009 843	9 923 720 515	4 263 298 536	5.4	0.16	0.49	11.546	63.65	3.604	3.309	0.644	1.338
deepsjeng	2 288 124 945 739	80 869 347 873	19 984 799 357	59 448 342 020	1 436 206 496	24.7	2.92	5.20	9.863	5.80	2.973	7.979	2.799	3.735
leela	2 625 129 460 409	80 675 242 597	2 803 257 082	74 618 676 357	3 253 309 159	3.5	0.38	5.68	6.479	18.20	4.290	0.044	1.767	1.282
namd	2 028 976 157 237	22 071 447 345	22 036 018 281	32 351 019	3 078 044	99.8	3.27	0.00	12.512	25.47	3.309	1.356	11.406	2.721
lbm	1 326 361 633 658	60 519 089 363	60 519 012 839	65 196	11 327	100.0	10.48	0.00	19.643	86.89	0.444	1.753	9.838	1.091
nab	1 734 299 363 064	43 335 663 671	28 093 888 918	13 442 916 323	1 798 858 430	64.8	5.20	1.55	7.987	40.00	1.573	2.822	0.989	1.158
xz_cld	401 325 607 575	6 541 500 109	2 739 318 343	3 420 468 756	381 713 010	41.9	3.38	1.70	11.838	31.57	3.973	1.386	3.233	5.036
xz_cpu2006docs	1 147 585 864 648	6 856 700 239	3 294 943 229	3 244 099 774	317 657 236	48.1	1.47	0.57	9.623	38.70	2.833	2.102	4.705	6.341
xz_combined	625 442 471 499	8 750 690 789	4 136 091 420	4 131 534 419	483 064 950	47.3	2.85	1.32	10.790	22.05	3.985	1.096	3.598	4.955
imagick	4 419 819 220 333	161 009 229 576	985 726 347	112 611 984 461	47 411 518 767	0.6	0.23	5.10	2.175	96.75	3.936	0.030	0.003	0.001
blender	1 882 360 450 846	37 346 363 686	1 777 349 126	31 450 061 421	4 118 953 139	4.8	0.38	3.34	8.221	17.84	2.763	2.064	2.294	5.233
povray	3 547 172 911 376	268 807 318 785	3 408 113 585	248 064 943 702	17 334 261 499	1.3	0.27	13.99	10.321	40.97	7.394	19.422	2.209	1.785
omnetpp	1 183 494 051 463	77 273 945 237	5 553 868	75 458 441 451	1 809 949 918	0.0	0.00	12.75	16.653	27.55	0.085	2.018	3.518	2.901
xalancbmk	1 280 238 262 454	32 984 432 390	514 678 757	30 661 938 211	1 807 815 421	1.6	0.13	4.79	17.454	68.78	10.520	15.819	3.098	5.031
x264_pass1	532 607 517 842	26 480 570 114	17 955 938 896	8 302 056 795	222 574 424	67.8	7.07	3.12	2.782	25.29	0.225	0.935	1.549	1.466
x264_pass2	2 060 055 578 301	88 359 230 989	60 166 511 652	24 461 154 321	3 731 565 017	68.1	6.25	2.37	2.284	31.33	0.359	0.870	1.391	1.418
x264_seek	1 966 389 483 167	85 688 434 921	58 863 030 064	23 684 590 688	3 140 814 169	68.7	6.41	2.41	2.305	27.65	0.320	0.973	1.282	1.409
gcc_pp_O3	224 853 966 645	17 984 502 435	877 592 244	16 879 105 802	227 804 388	4.9	1.23	15.01	9.964	37.45	2.052	5.060	1.067	2.035
gcc_pp_O2	266 929 823 668	21 831 478 671	1 077 827 255	20 499 147 410	254 504 006	4.9	1.25	15.36	10.052	34.58	2.160	5.085	1.120	1.988
gcc_smaller	254 671 373 628	17 265 308 635	931 220 362	16 129 309 979	204 778 295	5.4	1.09	12.67	11.466	43.86	2.779	4.427	0.962	1.871
gcc_ref32_O5	214 010 333 381	16 552 733 395	576 497 719	15 646 044 451	330 191 225	3.5	0.91	14.62	10.160	44.91	3.212	5.142	0.872	1.771
gcc_ref32_O3	318 165 395 272	17 589 262 663	566 719 818	16 818 061 154	204 481 691	3.2	0.68	10.57	11.324	68.76	3.920	6.151	1.006	2.063
mcf	888 079 703 023	7 407 299 183	3 229 620 454	4 144 174 494	33 504 235	43.6	1.40	0.93	22.054	72.44	7.929	7.606	3.444	5.730
perlbench_checkspam	1 309 521 527 045	120 008 085 121	23 137 601 197	93 974 994 957	2 895 488 967	19.3	7.98	14.35	4.538	16.93	0.845	1.777	0.971	0.621
perlbench_diffmail	778 566 201 895	66 166 562 175	3 721 969 520	61 702 382 195	742 210 461	5.6	1.96	15.85	6.043	30.54	2.425	7.792	0.390	1.565
perlbench_splitmail	736 991 491 190	79 002 095 413	7 227 608 106	70 427 685 127	1 346 802 180	9.1	4.16	19.11	9.965	7.65	0.397	4.282	1.510	0.827
bwaves_1	497 063 723 317	2 989 609 538	2 989 191 752	351 316	66 470	100.0	2.51	0.00	43.266	24.07	3.940	0.198	4.543	5.106
bwaves_2	784 691 829 793	4 892 060 128	4 891 557 364	404 109	98 655	100.0	2.59	0.00	42.951	24.39	3.803	0.230	7.461	5.187
bwaves_3	611 703 570 107	3 688 669 131	3 688 117 546	439 481	112 104	100.0	2.51	0.00	42.881	24.27	4.736	0.174	5.759	4.510
bwaves_4	745 048 195 941	4 606 093 801	4 605 716 550	319 055	58 197	100.0	2.56	0.00	42.814	24.15	3.885	0.165	4.857	5.803
exchange2	2 120 109 818 190	67 506 596 157	51 688 048 432	8 987 153 068	6 831 394 657	76.6	5.23	0.85	1.834	0.85	1.622	2.782	8.119	5.404
fotonik3d	2 457 798 604 788	19 150 475 290	18 811 665 225	309 878 797	28 931 268	98.2	3.36	0.03	16.671	41.15	6.136	0.171	0.106	3.443
roms	2 733 593 366 577	17 773 282 831	16 141 154 959	967 833 899	664 293 973	90.8	4.37	0.07	21.552	38.82	4.011	1.248	2.843	3.810
wrf	4 131 413 910 638	19 044 564 711	5 282 083 873	12 615 417 654	1 147 063 184	27.7	0.91	0.61	17.877	21.73	9.005	5.948	0.812	1.723
cam4	2 349 180 609 851	44 573 582 631	21 602 368 027	21 690 037 915	1 281 176 689	48.5	3.14	1.85	11.801	50.53	3.178	3.082	1.110	2.106
Total	58 391 814 127 468	1 959 776 643 990	**767 822 470 568**	1 084 245 719 802	107 708 453 623	39.2	3.78	3.71	0.000	—	—	—	—	—